

nnActive

A Framework for Evaluation of Active Learning in 3D Biomedical Segmentation

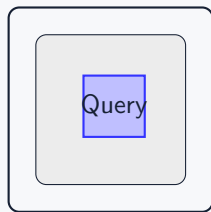
Carsten T. Lüth et al.

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Does active learning save annotation effort?

Why Active Learning for 3D Biomedical Segmentation?

- 3D biomedical segmentation needs accurate voxel-level labels.
- Expert annotation is expensive and often requires specialist knowledge.
- Active Learning asks: **which region should we annotate next?**
- The goal is not just higher Dice Score, but **less annotation effort**.



3D scan / patch

Annotate only the most useful regions

Why Is Better Evaluation Needed?

- Prior work gives **mixed conclusions** on whether Active Learning really beats Random sampling.
- A useful AL method should work **across datasets and annotation budgets**.
- It should still help when combined with other ways of using annotations efficiently (self- or semi-supervised learning).
- It should beat **realistic and inexpensive baselines**, not only naive Random sampling.
- Evaluation should measure realistically **annotation effort**, not only model performance.

Core question

Can we trust reported Active Learning gains in realistic 3D biomedical settings?

Four Pitfalls in Active Learning Evaluation

P1: Too Few Settings

Evaluation on too few datasets and annotation budgets, so result may not generalise.

P3: Weak Random Baseline

Uniform Random often selects mostly background, making Active Learning may look artificially strong

P2: Partial Annotations Used Poorly

2D training discards useful surrounding 3D context.

P4: Voxels $\neq$ Annotation Effort

Equal-size patches can require very different annotation effort.

How nnActive Addresses the Pitfalls

P1 → Broader Evaluation

4 biomedical datasets and 3 annotation-budget regimes.

P3 → Stronger Baselines

Introduces Foreground-Aware Random sampling.

P2 → Realistic 3D Training

3D nnU-Net trained with partial annotations and surrounding context.

P4 → Better Cost Metric

Introduces Foreground Efficiency (FG-Eff) as a proxy for annotation effort.